

Jaymin Chavda

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Education

Vishwakarma Government Engineering College	2023 – 2027
<i>Bachelor of Engineering in Computer Science & Engineering (Data Science)</i>	<i>CGPA: 8.04</i>
S.N. Vidhyalaya	2022 – 2023
<i>12th – GSEB (Science)</i>	<i>81.84%</i>
S.N. Vidhyalaya	2020 – 2021
<i>10th – GSEB</i>	<i>95.33%</i>

Projects

Rainfall Prediction using Machine Learning | *Python, scikit-learn, XGBoost, imbalanced-learn, Streamlit*

- Built an end-to-end binary classification pipeline to predict daily rainfall from weather data; engineered features to resolve multicollinearity and benchmarked Logistic Regression, Random Forest, and XGBoost, tuned with GridSearchCV on ROC-AUC.
- Addressed class imbalance using SMOTE and class weighting, tuned decision thresholds for precision/recall trade-offs, and deployed an interactive Streamlit dashboard delivering live predictions at 0.92 ROC-AUC.

Domain-Specific RAG System with Evaluation | *Python, BM25, Vector Search, Re-ranking, RAGAS*

- Built a question-answering system over a real, unstructured document corpus, comparing fixed-size, recursive, and semantic chunking strategies to select the best retrieval setup.
- Benchmarked OpenAI and open-source (BGE, E5) embedding models on cost and latency, and implemented hybrid search (BM25 + vector) with a re-ranking stage.
- Mitigated hallucinations using source citations, confidence thresholds, and an “I don’t know” fallback; evaluated faithfulness and answer relevance with RAGAS.

E-Commerce Sales Data Pipeline | *Python, Faker, Apache Kafka, Pandas, PostgreSQL, Apache Airflow, Streamlit*

- Built an end-to-end ETL pipeline (Python, Kafka, Pandas, PostgreSQL) for real-time e-commerce event processing.
- Automated pipeline orchestration using Apache Airflow DAGs and built an interactive Streamlit dashboard delivering live business insights: revenue trends, top-selling products, category performance, and regional sales analytics.

Technical Skills

Languages: Python, SQL

Data Analysis: Exploratory Data Analysis, Data Cleaning, Data Preprocessing, Feature Engineering, Pandas, NumPy

Machine Learning: scikit-learn, XGBoost, imbalanced-learn (SMOTE)

GenAI & RAG: RAG, Embeddings (OpenAI, BGE, E5), Hybrid Search (BM25 + Vector), Re-ranking, RAGAS

Data Engineering: ETL Pipelines, Apache Kafka, Apache Airflow

Databases: PostgreSQL, MySQL

Tools: Git, GitHub, GitHub Actions, Streamlit, Render

Certifications

Data Analysis with Python – freeCodeCamp	2024
Python for Data Science – IBM	2024
Python Certification – HackerRank	2025
Data Analytics Job Simulation – Deloitte Australia (Forage)	2026